TEXAS
The University of Texas at Austin

Origin and Function of Tuning Diversity in Macaque Visual Cortex

Goris, Simoncelli, Movshon (2015)

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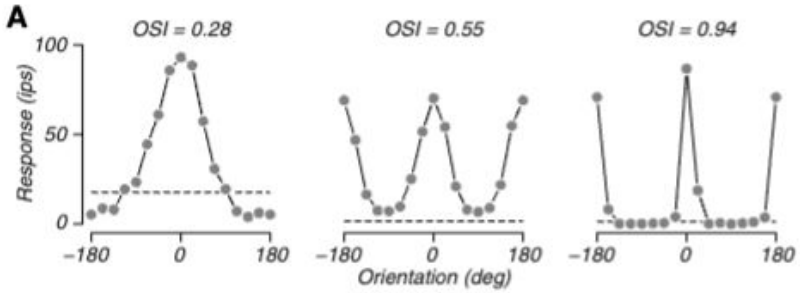
NEU 382V

Background: Orientation Selectivity

1. Neurons in early visual cortex (V1, V2) respond preferentially to edges at specific orientations

2. Each neuron has a tuning curve: response vs. orientation

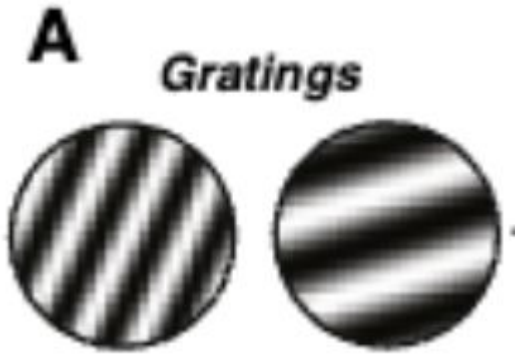
3. Tuning can be narrow (highly selective) or broad (less selective)



The Research Question

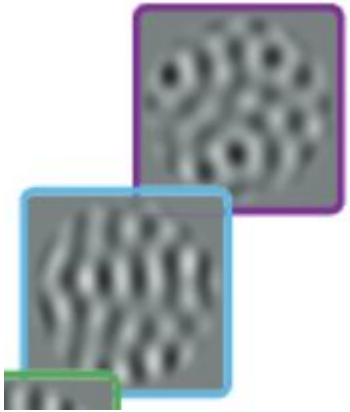
Beyond “how selectivity arises”

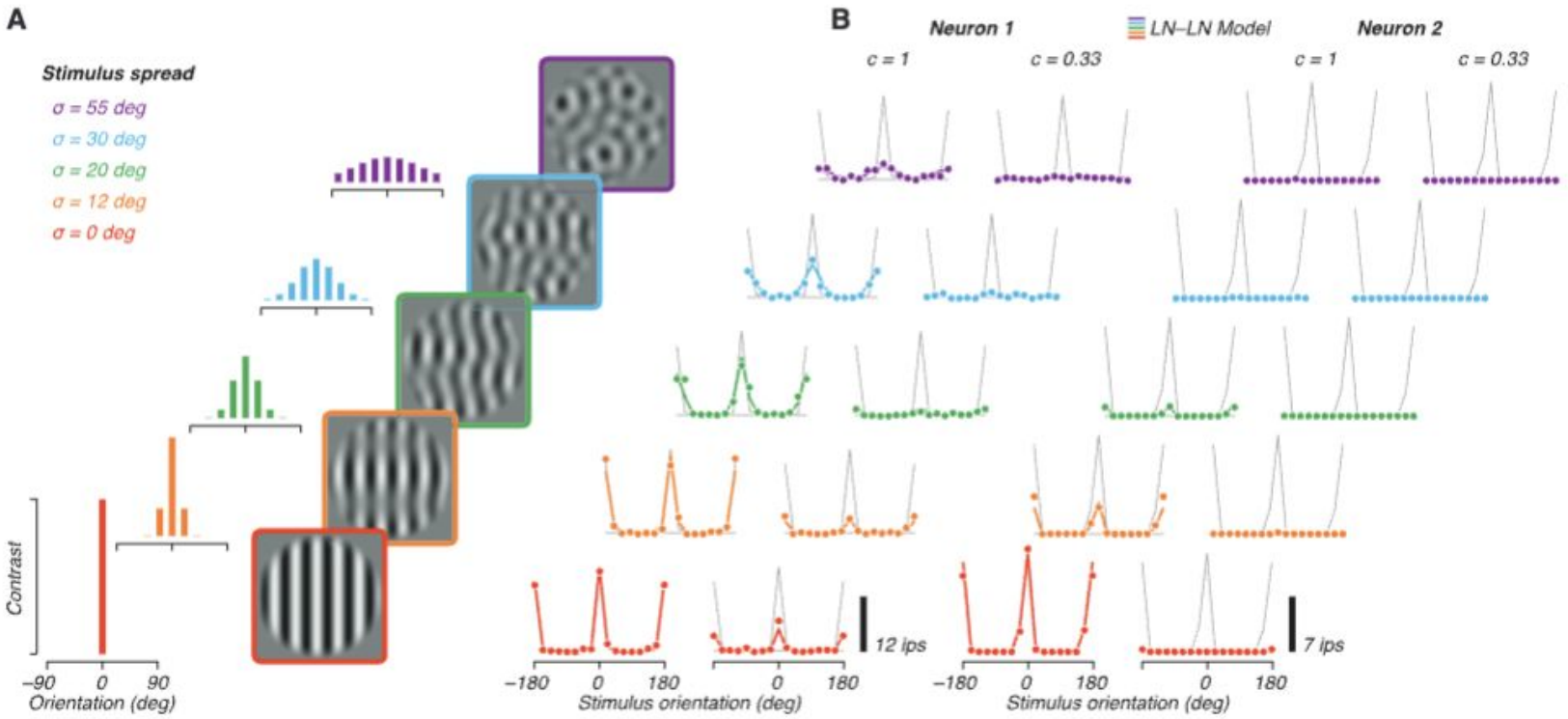
- What creates tuning diversity?
- What is this diversity for?



Recording & Stimuli

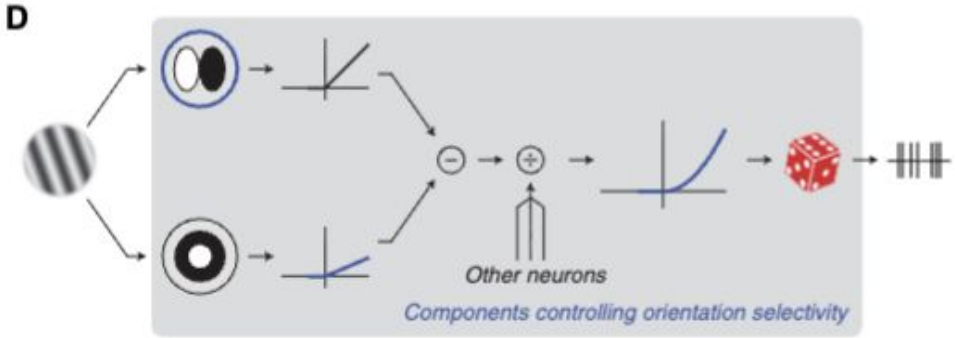
- V1 and V2 recordings in macaques
- Orientation mixtures, not just gratings
- Controlled “spread” of orientations




A
Stimulus spread
σ = 55 deg
σ = 30 deg
σ = 20 deg
σ = 12 deg
σ = 0 deg
Contrast
−90
0
90
Orientation (deg)
B
Neuron 1
LN–LN Model
Neuron 2
c = 1
c = 0.33
c = 1
c = 0.33
12 ips
7 ips
−180
0
180
Stimulus orientation (deg)
−180
0
180
Stimulus orientation (deg)

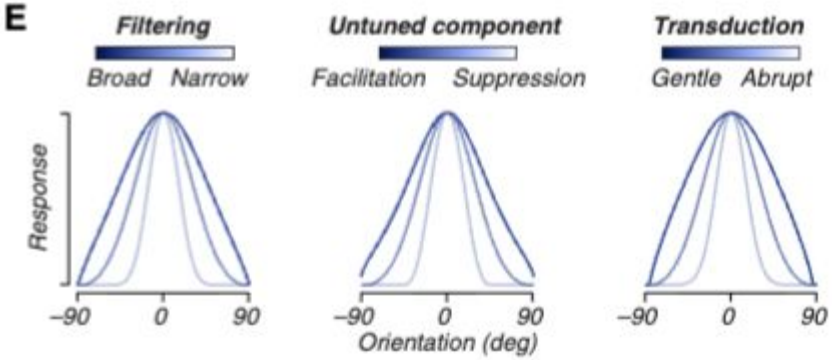
The LN–LN Cascade Model

- Two-channel model
- Oriented channel + untuned channel
- Filtering → normalization → nonlinearity



Three Core Mechanisms

- Filtering
- Suppression / facilitation
- Transduction nonlinearity



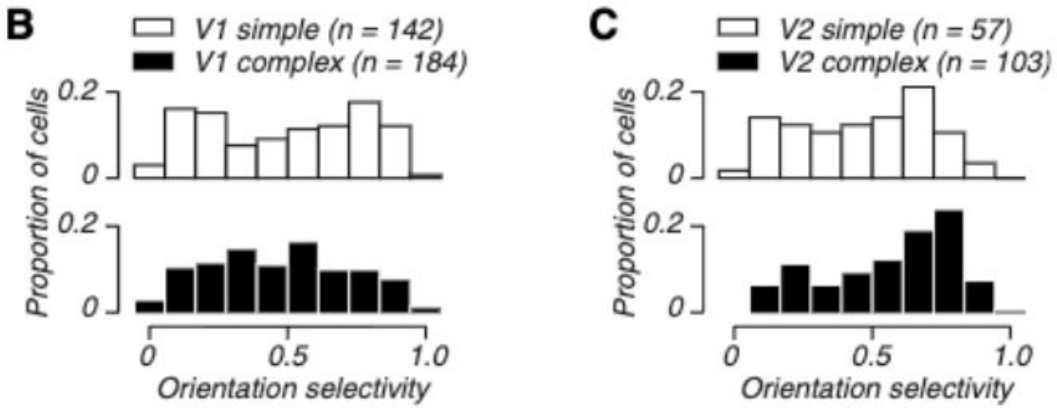
What Drives Diversity?

- Model explains most diversity
- Filtering dominates
- Suppression & nonlinearity refine

V1 vs V2 Differences

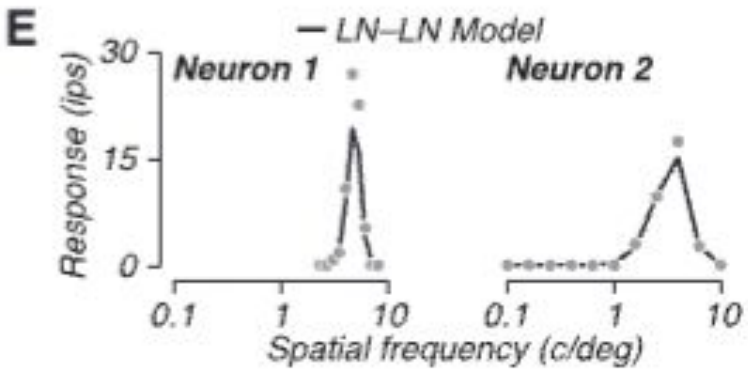
V1: more untuned excitation

V2: more untuned inhibition



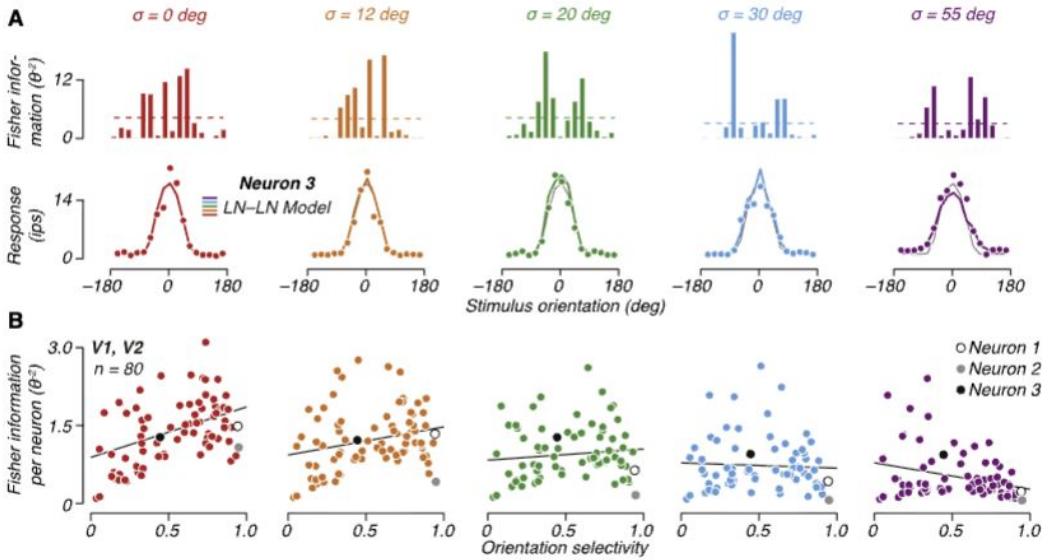
Spatial Frequency Tuning

- Same model explains spatial frequency diversity
- Shared computational origin



Specialists vs Generalists

- Narrow = specialists
- Broad = generalists



The Natural World Is Diverse

- Natural images contain orientation mixtures
- Matches tuning diversity

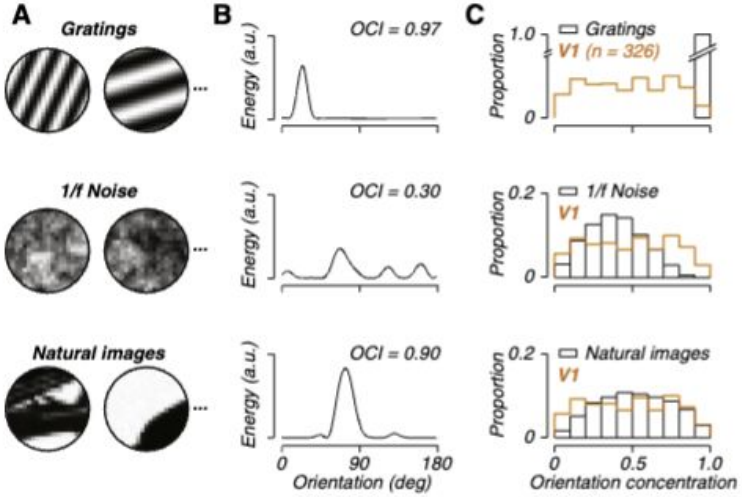


Figure 7. Distribution of Local Orientations for Three Image Ensembles

Coding Advantage

Diverse populations outperform uniform ones

Especially for natural images

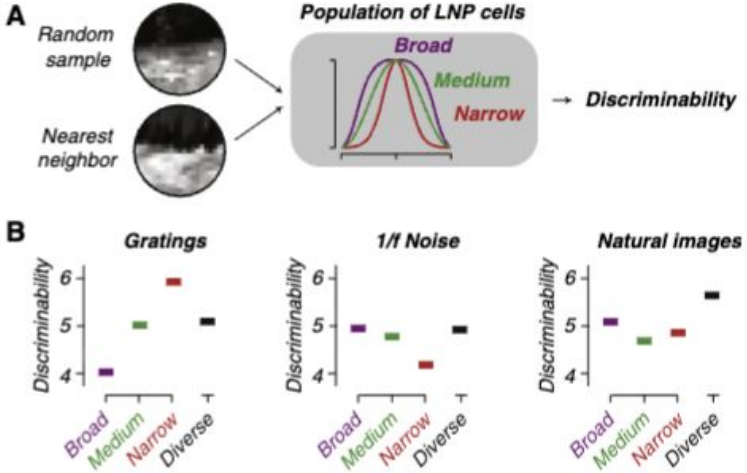


Figure 8. Tuning Diversity Is Beneficial for Visual Coding

Summary/Key Takeaways

- Origin: filtering variability
- Function: efficient encoding

Discussion

- Do species with different visual environments show the same distributions of tuning diversity?
- How might these ideas translate to artificial vision models (e.g., CNNs)
- Is tuning diversity learned through visual experience, or largely specified during development?